

COMPARISON WORKFILE

1 Tonne Mobile Jib Crane, Structural Verification

Rev 1 and Rev 2 compared: reports and MATLAB scripts

Item	Detail
Project reference	40467361
Documents compared	Rev 1 (08 Jun 2026, 22 pp) and Rev 2 (09 Jun 2026, 27 pp)
Scripts compared	jib_crane_verification_v1.m (1031 lines) and v2.m (1197 lines)
Configuration	Wheels only, outriggers stowed
Analysis tool	MATLAB R2026a (parametric; hand calc plus in-script frame FEA)
Prepared	09 June 2026

None of the safety factors moved between the two revisions. The governing case, the checks that fail, and the overall conclusion are the same in both. Rev 2 is about explanation, not new results. It sets out why the crane passed a 150% proof-load test even though the column works out at first yield in the elastic check, and it supports that with three separate cross-checks. It is also clearer about which inputs still have to be confirmed on the actual machine.

1. Purpose of this Workfile

Three questions are covered here. First, what does the second report and the v2 script add that the first did not. Second, how do the two MATLAB scripts differ. Third, how do the two reports differ. Both revisions use the same geometry, the same load model and the same fourteen checks, and nothing in the checks changed. Rev 2 adds interpretation and traceability on top of results that stayed put.

2. The Engineering Verdict Did Not Change

It helps to set down first what did not change, so the new material reads in context. All of the following carry over unchanged from Rev 1 to Rev 2:

Quantity (unchanged in both)	Value	Verdict
Governing case	700 kg at 2320 mm reach	Mid-reach governs
Governing design moment	19.91 kN·m	
Column, combined axial + bending	277 MPa, FoS 0.99	Fail (at yield)
Column-to-base weld	235 MPa, FoS 0.85	Fail
Stability (worst point)	FoS 1.34	Fail
Boom outer section	186 MPa, FoS 1.48	Marginal
Outrigger arm	200 MPa, FoS 1.38	Fail
Cylinder force / pivot reaction	F _{cyl} 57.49 kN / F _{pivot} 63.85 kN	
Wheel reactions	Front 15.16 kN / Rear 0.29 kN	Near tipping
Member / weld allowable	183 MPa / 200.7 MPa	
Checks below target	8 of 14	Cannot certify to chart

A note on the cylinder force. Both scripts work out the luffing-cylinder lever arm the same way, $a_{cyl} = cyl_anchor_boom \times \sin(60^\circ) = 0.346$ m, which gives $F_{cyl} = M / a_{cyl} = 57.49$ kN. Rev 2 does not change this. It prints the lever arm and marks it for checking on site, nothing more.

3. What Rev 2 / v2 Adds

Five items are present in Rev 2 and the v2 script but absent from the first set. They are given roughly in order of importance.

3.1 The 150% Proof-Load Reconciliation (Report §12 / script TEST 9)

This is the largest addition and the main reason for the revision. There is an obvious tension in Rev 1 that it never deals with. The elastic check puts the column at FoS 0.99, which is first-fibre yield at the rated dynamic load, yet the crane has passed a static 150% proof-load test. Rev 2 explains how both can be true.

The reason is the limit state being checked. The report's checks are elastic working-stress checks: peak elastic stress against first yield, with a 1.5 margin asked for. A proof test is a different measure. It shows

ultimate capacity, and for a steel section that is set by the plastic moment $M_p = Z_x \times f_y$, not by first yield. A compact hollow section keeps reserve between first yield and a full plastic hinge equal to its shape factor. To put numbers on this, v2 adds the plastic section modulus Z_x to the section properties; v1 carried only A , I_x and S_x .

Column bending-moment ladder (new in Rev 2):

Quantity (column RHS 80×120×6)	Value	Consequence
Elastic modulus S_x / plastic modulus Z_x	73.0 cm ³ / 89.7 cm ³	Shape factor 1.23
Applied moment at 1.25×SWL	19.91 kN·m	Elastic FoS 0.99
First-yield moment M_y (nominal 275)	20.08 kN·m	Yield just reached
Applied moment at 150% static proof	23.90 kN·m	Demand in the test
Plastic moment M_p (nominal 275)	24.67 kN·m	Survives, +3% margin
Plastic moment M_p (mill 320)	28.71 kN·m	Survives, +20% margin

The other members at the proof load are reported too, also new. At 150% the boom outer section reaches 223 MPa and the outrigger arm 240 MPa, both still below the 275 MPa yield. So the column is the only member close to its limit, which is why nothing else showed distress in the physical test.

Three further points work in the crane's favour and are not credited in the elastic check. The proof test is static, so it carries no 1.25 dynamic factor. Real S275 mill steel usually returns around 320 MPa rather than the nominal 275. And the luffing cylinder acts partly as a diagonal strut, taking some of the column bending that the cantilever model puts entirely into the column.

Reframed conclusion (Rev 2 only): The 0.99 figure is a conservative first-yield result, not a forecast of failure. The structure can carry 150% once, and the test proves it. What it lacks, on the as-built numbers, is the usual 1.5 working-stress margin at mid-reach for repeated lifting, and working at extreme-fibre yield every day is a fatigue concern. The report is conservative, not wrong.

What changed in the script: v2 adds TEST 9, two new inputs (f_y _actual = 320 MPa and proof_factor = 1.50), the plastic modulus Z_x in the section-property function, two helper functions (proof_txt and elastic_txt), and the proof variables in the saved workspace.

3.2 Hydraulic-Pressure Cross-Check (Report §6.1 / script TEST 4)

Rev 1 gave only the cylinder force. Rev 2 adds a pressure check that ties that force back to the hydraulics. It also shows the proof test really did load the structure, rather than the relief valve cutting in first.

Hydraulic cross-check (luffing cylinder, 50 mm bore, $A = 1963 \text{ mm}^2$)	Pressure	Reading
Pressure to hold 1.25×SWL (57.49 kN / 1963 mm ²)	29.3 MPa (293 bar)	Within normal duty
Pressure at 150% proof load	35.1 MPa (351 bar)	Relief must pass it; hydraulics did not cap the proof

Hydraulic cross-check (luffing cylinder, 50 mm bore, A = 1963 mm ²)	Pressure	Reading
Pressure if bore were only 30 mm	≈ 813 bar	Not plausible, so 50 mm is the luffer

Rev 2 concludes that the relief valve had to pass about 351 bar during the proof test, so it was the structure that carried the full 150%, not the hydraulics. This backs up the proof-load reconciliation in 3.1.

3.3 Cylinder Rod-End Pin Cross-Check at Base Detail-4 (Report §7.4 / script TEST 4)

Rev 1 modelled the cylinder mount pin at Ø20, got FoS 1.20, and called it the genuine weak pin, undersized on any basis. Rev 2 looks again and makes the verdict conditional. The mount pin is not dimensioned on the cylinder views. The only Ø40 hinge pin near the mount is Base Detail-4 (shank Ø40 × 180, head Ø52). A Ø20 pin taking 57 kN does not add up; it would be smaller than the Ø30 boom-pivot pin. At Ø40 the pin passes with room to spare.

Cylinder mount pin	Shear stress	Bearing	FoS	Verdict
Ø20 mm (Rev 1, as modelled)	91.5 MPa	191.6 MPa	1.20	Fail
Ø40 mm (Rev 2, Detail-4)	22.9 MPa	95.8 MPa	4.81	Pass

The effect is that one of Rev 1's eight failures is probably just a misread dimension. The summary table keeps the conservative Ø20 value so it stays consistent with the issued numbers, but the action changes from replace the pin to confirm the fitted diameter. In the code, v2 adds `pin.d_cyl_alt = 0.040` and prints the Detail-4 result next to the Ø20 one.

3.4 Corrected Cylinder Identification and a Drawing-Verification Register

v2 fixes a misread in the cylinder definitions. v1 took “Type 80 / Type 50” to mean the bore, 80 mm and 50 mm. v2 reads these as series names and takes the drawing's “core diameter” as the true bore:

Cylinder	v1 reading	v2 (corrected) reading
Type 80 (luffing)	bore 80 mm, rod 50 mm, stroke 700	bore 50 mm, stroke 700; luffing
Type 50 (telescoping)	bore 50 mm, rod 30 mm, stroke 775	bore 30 mm, stroke 775; telescoping

The fix does not change the cylinder force, which comes from boom statics and does not depend on bore. But it is needed for the pressure check in 3.2 to mean anything, and it settles which cylinder does the luffing. v2 also adds a drawing-verification register at the top of the input block. It records, callout by callout, what the drawings confirm (assembly dimensions, the warning plate, the ARM1 and BASE details) and what they cannot settle, flagged ESTIMATE or VERIFY (the cylinder anchor coordinates and lever arm, the cylinder pin diameter, the counterweight CG and the wheel positions). For completeness it notes some non-structural items now read off the drawings: the slew bearing (OD Ø210, bore Ø125, plate 180×180), the slew drive (module 2.5, 22-tooth pinion, 98-tooth ring), the column base plate (Ø240, T7.5, hub Ø85) and the support-leg lengths (750 / 798 mm).

3.5 Warning-Plate 420 mm Stability Reconciliation (Report §14 / script base block)

v2 brings out a stability uncertainty that Rev 1 left alone. The 420 mm front dimension on the warning plate only fits the load at the shortest reach (1400 mm) sitting about 420 mm past the front tipping line if the front wheel is roughly 980 mm ahead of the column. With the 1200 mm value used in the analysis, the load sits only about 200 mm past it. The two readings bracket the stability result, so the front overhang (L_{fwd}) and the centre-of-gravity offset (e_{cg}) need to be measured, and the crane weighed, before sign-off. Rev 2 adds this to the items to confirm.

4. Differences Between the Two MATLAB Scripts

v2 is 166 lines longer than v1 (1197 against 1031). A diff shows 193 lines added and 27 changed. The table below lists what differs. None of the original calculations were touched.

Aspect	v1	v2
Analysis blocks	TEST 1–8	TEST 1–9 (adds proof-load reconciliation)
New inputs	none	fy_actual = 320 MPa; proof_factor = 1.50; pin.d_cyl_alt = Ø40; slew bearing/gear; column base plate; leg lengths
Section properties	A, Ix, Sx	plus Zx (plastic section modulus)
Cylinder structs	cyl_lift (Type 80, 80 mm) / cyl_luff (Type 50, 50 mm)	cyl_luff (Type 80, 50 mm) and cyl_tele (Type 50, 30 mm); corrected
New computations	none	hydraulic pressure (required, at proof, 30 mm alternative); Detail-4 Ø40 pin; plastic-moment ladder; member stresses at proof
Helper functions	baseline set	plus proof_txt and elastic_txt
Comments / flags	standard read-off comments	drawing-verification register; VERIFY flags on L_fwd, L_rwd, lever arm, pin; 420 mm stability note
Saved workspace	baseline variables	plus My/Mp (nominal and actual), M_applied, M_proof, proof_factor, fy_actual, F_cyl, pressures, member proof stresses
F_cyl, F_pivot, wheel reactions	baseline	identical, unchanged
Weld allowable in code	$0.7 \times f_u / 1.5$	$0.7 \times f_u / 1.5$, identical
14-check FoS results	baseline	identical, unchanged

5. Differences Between the Two Reports

Rev 2 runs to 27 pages and 15 sections, up from 22 and 14. The new Section 12 on the proof-load reconciliation moves Conclusions to §13, Assumptions to §14 and Standards to §15. Section by section:

Report area	Rev 1	Rev 2
New section	none	§12 Reconciliation with the 150% Proof-Load Test
§6.1 Cylinder / pivot	force only	adds the hydraulic-pressure cross-check; lever arm a_cyl flagged for site confirmation
§7.4 Pins	Ø20 cylinder pin called the genuine weak pin	adds the Detail-4 Ø40 cross-check (FoS 4.81); verdict made conditional on the as-built size

Report area	Rev 1	Rev 2
§3.1 Weld allowable text	“ $0.5 \times f_y / 1.5 \dots = 200.7$ ” (wording inconsistent with the code)	“ $0.7 \times f_u / 1.5$, $f_u = 430$ ” (matches the code; same 200.7 / 183 numbers)
Executive summary	verdict only	adds a paragraph reconciling the proof test with the elastic findings
Document control	Rev 0–1 history	adds a proof-test basis bullet and the Rev 2 history line
§14 Items to confirm	self-weight, CG, overhang, welds, pin basis	adds cylinder lever arm and pin, mill yield, the warning-plate 420 mm reconciliation
Cylinder-pin language	“undersized on any basis”	“undersized if Ø20; passes at 4.81 if the Ø40 Detail-4 pin”
14-check FoS table and governing results	baseline	identical, unchanged

6. Summary

Rev 2 and v2 change no result. Their value is in the explanation and the evidence behind it. Rev 1 set two facts side by side without joining them: the column works out at yield, so it fails the elastic check, yet the crane passed a 150% proof test. Rev 2 joins them through the elastic-versus-plastic limit state and puts numbers on the plastic reserve, a shape factor of 1.23, with M_p clearing the proof demand by 3% on nominal steel and 20% on realistic mill steel.

Three cross-checks support this, and they agree. The pressure check shows the test loaded the structure rather than the relief valve. The Detail-4 Ø40 pin shows the cylinder-pin failure is most likely a misread dimension. The mill-yield sensitivity shows real steel adds margin. Rev 2 then states plainly what still needs confirming on the machine: the cylinder pin diameter, the cylinder lever arm, the wheel and CG geometry, and the actual mill yield. The conclusion is the same as Rev 1, but in Rev 2 it is defensible, and one of the original eight failures has largely been explained.

End of comparison. 1 Tonne Mobile Jib Crane, Rev 1 and Rev 2. Confidential, issued for review.